Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/02/2004
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Project Title:	EXPLORATION AND DEVELOPMENT OF A NOVEL MICROPROPULSION TECHNIQUE FOR NAVIGATION IN LOW GRAVITY ENVIRONMENTS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20373-001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gamezo, Vadim N. (Naval Research Laboratory)		
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	This work is based on a micropropulsion idea generated from a relatively recent discovery concerning the behavior of laminar flames propagating in very narrow tubes. When a slow laminar flame is ignited near the closed end of a tube, the material ahead of the flame is accelerated and a boundary layer is formed along the walls in the unreacted material. Due to the presence of the boundary layer, the flow velocity changes across the channel, increasing from zero at the wall to a maximum in the middle of the tube. Previous simulations have shown that the nonuniform flow stretches the flame, so that the shape of the flame becomes similar to the velocity profile. This increases the flame surface area, thus accelerating the energy generation and the flow. There is little change in the laminar burning rate, but the speed of the flame that propagates with the flow grows very rapidly in the laboratory frame of reference, from centimeters per second to hundreds of meters per second for adiabatic walls.		

Task Description:	The high-speed flow created by the accelerating flame has the potential to provide a significant thrust that can be further enhanced by attaching an appropriately shaped exist nozzle. The thrust will gradually increase as the flame and flow accelerate, and reach a maximum when the flame approaches the open end of the tube or the nozzle. We consider regimes in which the flame remains laminar until the end of the tube and does not produce strong shocks or detonations. In contrast to the majority of other combustion-based propulsion concepts, most of the thrust here can be provided by the material accelerated ahead of the flame. Therefore, a propulsion device could be only partially filled with a combustible mixture. The rest of the material can be an inert gas or combustion products from a previous pulse. To explore the possibility of using flame acceleration in narrow tubes for micropropulsion. This micropropulsion concept is of particular interest for low-gravity environments, where relatively small, controlled thrusts could be used for navigational corrections. In such cases, there might not be a need for the high repetition rates required for pulse combustion and pulse detonation engines.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A laminar flame propagating towards the open end of a narrow channel filled with a gaseous combustible mixture can accelerate or oscillate, depending on the wall temperature and the channel width. The accelerating flame is able to produce a high-speed flow that has the potential to provide significant thrust, and this can be used in micropropulsion devices. We study these phenomena using multidimensional reactive Navier-Stokes numerical simulations, and show that for adiabatic walls, the maximum flame acceleration occurs when the the channel is about 10 times larger than the reaction zone of a laminar flame. In three-dimensional square channels, the flame speed increases roughly 2 times faster than in idealized two-dimensional channels. The accelerating flame generates weak compression waves that propagate with a local sound speed ahead of the flame. If the channel is long enough, the compression waves may converge to a strong shock, and eventually trigger a detonation that develops in the shock-compressed material near the wall. In shorter channels, the flame and the unreacted material ahead of it can accelerate to the velocities close to the sound speed without creating strong shocks. This combustion regime is of particular interest for micropropulsion because it allows an efficient use of fuel and a gradual development of the trust. The results of numerical simulations were used to determine key specifications for proof-of-principle experiments at NASA Glenn Research Center.
Bibliography Type:	Description: (Last Updated: 07/20/2007)
Abstracts for Journals and Proceedings	Gamezo, V. N.; Oran, E. S. "Flame Acceleration and DDT in Narrow Tubes." 57th Meeting of the Division of Fluid Dynamics of the American Physical Society, Seattle, Washington, November 21-23, 2004. NK.002 , Nov-2004
Abstracts for Journals and Proceedings	Gamezo, V. N.; Oran, E. S. "Flame Acceleration in Narrow Tubes: Applications for Micropropulsion in Low-Gravity Environments." 43th AIAA Aerospace Sciences Meeting & Exhibit, Reno, Nevada, January 10-13, 2005 AiAA-2005- (accepted) , Jan-2005